

|                     |   |                           |
|---------------------|---|---------------------------|
| unsigned (int)      | 2 | 0 to 65535                |
| (signed) long (int) | 4 | -2147483648 to 2147483647 |
| unsigned long (int) | 4 | 0 to 4294967295           |

### Floating-point types

Types float and double, together with the long double variant, are considered floating-point types. CCS's implementation of ANSI Standard considers all three to be the same type.

Floating point in CCS is implemented using the Microchip AN575 32-bit format (IEEE 754 compliant).

The table below is the overview of the floating-point types:

| Type        | Size in bytes | Range                                                                   |
|-------------|---------------|-------------------------------------------------------------------------|
| float       | 4             | $\pm 1.17549435082 \times 10^{-38}$ to $\pm 6.80564774407 \times 10^38$ |
| double      | 4             | $\pm 1.17549435082 \times 10^{-38}$ to $\pm 6.80564774407 \times 10^38$ |
| long double | 4             | $\pm 1.17549435082 \times 10^{-38}$ to $\pm 6.80564774407 \times 10^38$ |

### Enumerations

An enumeration data type is used for representing an abstract, discreet set of values with appropriate symbolic names.

#### Enumeration Declaration

Enumeration is declared like this:

```
enumtag {enumeration-list};
```

Here, tag is an optional name of the enumeration; enumeration-list is a comma-delimited list of discreet values, enumerators (or enumeration constants). Each enumerator is assigned a fixed integral value. In the absence of explicit initializers, the first enumerator is set to zero, and each succeeding enumerator is set to one more than its predecessor.

Variables of enum type are declared same as variables of any other type. For example, the following declaration:

```
enumcolors { black, red, green, blue, violet, white } c;
```

establishes a unique integral type, enumcolors, a variable c of this type, and a set of enumerators with constant integer values (black = 0, red = 1, ...). In C, a variable of an enumerated type can be assigned any value of type int – no type checking beyond that is enforced. That is: